Brazil State-Size Fiscal Model: Review and Assessment

Review of F. A. Moreira (25 October 2026), *Reducing, Holding and Expanding the Size of the State: A Modular Stochastic Framework for Evaluating Fiscal Reform Scenarios in Brazil*. Written in simplified technical English (ASD-STE100 style).

Part 1. Summary of the paper

1. Objective

- The paper builds a tool that shows what happens when Brazil cuts, holds, or increases public spending.
- The time span is 5 to 20 years.
- The tool compares each scenario with a status-quo baseline (S0).
- The tool does not make a forecast. It does not rank the two 2026 candidates.
- The author's main claim is about method: every parameter has a source, and the model reports its own failures.

2. Economic model

The model is **not** a DSGE model. It is a modular accounting and comparative-statics model with Monte Carlo simulation. It has nine linked blocks:

1. **Demographics and fiscal baseline.** Age profiles from National Transfer Accounts calculate pension and health costs as the population ages.
2. **Macro block.** A Cobb-Douglas production function, $Y = A \cdot K^{\alpha} \cdot (hL)^{1-\alpha}$, calculates growth. The standard debt identity calculates debt.
3. **Fiscal multipliers.** Brazil-specific multipliers for each spending category (Sanches et al.) convert spending changes into GDP effects.
4. **Household microsimulation.** This block uses PNAD 2025 data and is static. It calculates the Gini coefficient, poverty, and income shares.
5. **Social outcomes.** Health spending changes infant mortality and life expectancy through fixed elasticities.
6. **Long-run feedback.** This block covers schooling, health, public capital (off), and a debt-linked risk premium.
7. **External block.** This block covers commodity prices and the exchange rate. Each draw applies the same shock to all scenarios.
8. **Governance.** This block covers routine leakage and rare large corruption shocks. All scenarios get the same values.
9. **Monte Carlo and back-test.** The model runs 1,000 draws and checks the results against Brazil's history.

3. Key assumptions

- **No behavioural response.** Prices, wages, labour supply, and monetary policy do not react, except through the named channels.
- **Fixed employment rate** (2019 level), with an Okun-law adjustment only.
- **Revenue is a constant share of GDP** (elasticity of about 1).
- **Real interest rate:** the median is 5.05%, from 2021–2025 data.
- **TFP growth** comes from Penn World Table data. This gives potential growth of 0.94%/year. The IMF uses 2.5%.

- **Parameters are independent** in the Monte Carlo. The model samples them from triangular or uniform ranges.
- **Health elasticities are symmetric.** Spending cuts and increases have the same effect.
- **Risk premium is linear:** 5 bp per pp of debt, with a cap.
- **Default cut package:** 40% pensions, 25% subsidies, 20% efficiency, 15% other. Investment and targeted transfers are protected.
- **Candidate programmes:** the author had to estimate many details that the programmes do not specify. Both programmes also get a +0.25 pp delivery slippage.

4. Key takeaways

1. **Composition and persistence are as important as size.** A 4 pp cut costs less than 1% of GDP with transitory multipliers. It costs about 3.9% with persistent multipliers.
2. **Ageing alone adds cost.** Age-related spending increases by 3.3 pp of GDP in 10 years and 7.3 pp in 20 years, with no policy change.
3. **Debt rises fast under the status quo.** Debt goes from 78.6% to about 145% of GDP by 2035. The interest-growth gap causes 61% of this increase, not the primary deficit.
4. **Cuts reduce debt but increase inequality.** An ambitious cut reduces debt by about 23 pp. It also increases the Gini coefficient and infant mortality a little. Expansions have the opposite effect.
5. **The model fails its own absolute-debt test.** It overshoots 2024 debt by 12.5–16 pp. The main cause is discretionary Treasury–BNDES–Central Bank operations. The author therefore publishes only differences against the baseline. He widens the uncertainty bands and does not add a fitted correction.
6. **The two 2026 programmes** are two different "closure rules" for the same debt equation. The paper does not publish a comparison of them.

5. Points to watch

- Most of the multiplier calibration comes from one paper that was not yet published.
- Brazil's negative fiscal-reaction coefficient (−0.037) is an in-house estimate. Nobody else has reviewed it.
- The low growth baseline (0.94%) makes debt results worse than IMF-based results.

Part 2. Assessment against four questions

> **The Keynesian benchmark.** Keynes proposed **countercyclical** fiscal policy. The government increases spending and accepts deficits **during a downturn**, when private demand is weak and idle resources are available. The government then reduces spending and builds surpluses **during the boom**, to pay back the debt and to prevent overheating. The second half of this rule is equally important. This assessment uses this benchmark.

1. Does the model consider the economic cycle?

No, not in practice.

- The model has no **output gap**. It calculates only potential growth and an Okun-law unemployment adjustment.
- The multipliers have one value per spending category. They do **not** change with the cycle.
- The paper cites the evidence that multipliers are larger in recessions (Auerbach–Gorodnichenko, and Pires for Brazil). It does **not** use this evidence in the model.
- The only time dimension is "transitory vs persistent" multipliers. That is about how long an effect lasts. It is not about when the spending happens.

Result: In this model, a 1 pp spending increase has the same effect in a boom and in a recession. That is the main gap relative to Keynesian theory.

2. Can it identify spending at the wrong time?

No.

- The model is forward-looking. It runs scenarios. It does not analyse past fiscal decisions.
- The back-test checks only whether the debt equation reproduces history. It does not ask whether past spending was **procyclical**.
- The back-test does find two clues, but it does not explain them: (a) the 2011–2022 Treasury–BNDES operations are the largest unexplained debt residual; (b) no specification explains the 2014–2016 recession.
- These are exactly the periods where a timing analysis would be useful. The 2011–2014 period had fiscal expansion while growth was slowing and inflation was high.

3. Can it show that spending bought back external debt?

No.

- The model uses **gross general-government debt** as one number. It has no split by currency, by holder (domestic or foreign), or by maturity.
- The exchange rate affects debt only through inflation pass-through. It does **not** revalue foreign-currency debt.
- The only tool for a debt operation is the residual term a_t, which the model uses for privatisation proceeds.
- **Context:** Most of Brazil's public debt is domestic and in reais. Large FX reserves also make the public sector a net external creditor. So the main risk today is **domestic debt at a high Selic rate**, not external debt. The model cannot show this composition.

4. Long-term effects of continuous spending on rates, FX, and inequality

The transmission chain is valid:

> More spending → GDP up at first → higher debt and inflation risk → Central Bank raises the Selic → the real weakens → imported inflation → real wages fall → growth slows.

The model captures only a small part of this chain:

Link in the chain	In the model?
Spending increases GDP at first	Yes (multipliers)
Debt increases the risk premium	Partly: linear, 5 bp per pp, with a cap
Central Bank raises the Selic in response	**No.** The paper says there is no monetary reaction function.
Fiscal stress weakens the real	**No.** FX shocks are external and the same for all scenarios.
Higher rates crowd out private investment	**No**
Inflation reduces the real income of the poor	**No**
High interest goes to bondholders, who are mostly in the top income groups	**No.** The microsimulation does not model interest income.
Higher inequality slows growth	Specified but **switched off**

Consequence: The model reports that expansions **reduce** the Gini coefficient. This result is probably biased in favour of expansion. The model omits the regressive channels: the inflation tax, high interest payments to the rich, and the cost of a weaker currency. The model sees the short-term benefit and not the long-term cost.

Part 3. How to improve the model

1. **Add the cycle.** Model an output gap. Use state-dependent multipliers, for example the Markov-switching estimates for Brazil.
2. **Add a fiscal timing diagnostic.** Calculate the structural primary balance and the fiscal impulse each year. Compare them with the output gap. This marks procyclical years (for example 2011–2014).
3. **Add a monetary reaction function.** A Taylor rule for the Selic that responds to inflation and fiscal expectations.
4. **Make the exchange rate endogenous.** Link FX to the fiscal path and the risk premium, not only to external shocks.
5. **Split the debt.** Model it by currency, indexation (Selic-linked, inflation-linked, fixed), maturity, and holder. Include FX reserves.
6. **Add the distribution of interest income and inflation** to the microsimulation. This shows who gains from high rates and who pays the inflation tax.
7. **Switch on the inequality-to-growth feedback.** Connect the Gini path to growth.
8. **Make the risk premium nonlinear by default.** The paper's own fiscal-fatigue variant shows how fast debt can explode.

> **Summary:** The model is honest and well documented. But it treats fiscal policy as **timeless and isolated from monetary policy and FX**. In Brazil, the cycle, the Selic, and the real are the main transmission channels, so the long-term results probably favour spending too much.